

Preprocessing Project

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Part 1: Data Loading and Initial Exploration

1.1) DATASET OVERVIEW

Dimensions

We found that the data had 1,010 rows and 16 columns, as shown by the results of

`print(df.shape())` below:

```
(1010, 16)
```

Sample of first 5 rows

Using `print(df.head())` to get an idea what the raw data looks like yields the following rows:

	Unnamed: 0	Music	Techno	Movies	History	...	Finances	Age	Siblings	Gender	Village - town
0	0	5.0	1.0	5.0	1.0	...	3.0	20.0	1.0	female	village
1	1	4.0	1.0	5.0	1.0	...	3.0	19.0	2.0	female	city
2	2	5.0	1.0	5.0	1.0	...	2.0	20.0	2.0	female	city
3	3	5.0	2.0	5.0	4.0	...	2.0	22.0	1.0	female	city
4	4	5.0	2.0	5.0	3.0	...	4.0	20.0	1.0	female	village

Summary of data types:

Disregarding the “Non-Null Count” column, the information provided by running `print(df.info())` gave us a good summary by column of the corresponding data types (labeled as “Dtype”).

#	Column	Non-Null Count	Dtype
0	Unnamed: 0	1010 non-null	int64
1	Music	1007 non-null	float64
2	Techno	1003 non-null	float64
3	Movies	1004 non-null	float64
4	History	1008 non-null	float64
5	Mathematics	1007 non-null	float64
6	Pets	1006 non-null	float64
7	Spiders	1005 non-null	float64
8	Loneliness	1009 non-null	float64
9	Parents' advice	1008 non-null	float64
10	Internet usage	1010 non-null	str
11	Finances	1007 non-null	float64
12	Age	1003 non-null	float64
13	Siblings	1004 non-null	float64
14	Gender	1004 non-null	str
15	Village - town	1006 non-null	str
dtypes: float64(12), int64(1), str(3)			

1.2) INITIAL DATA QUALITY

Target variable:

Since the dataset is being prepared to predict loneliness, we can identify our target variable as being `Loneliness`. Following is a graph showing the distribution of the Loneliness variable.

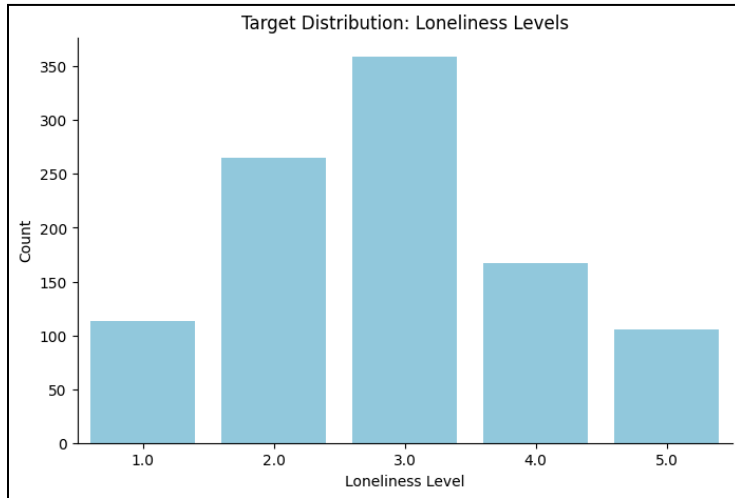


Figure 1: Target Distribution of Loneliness. The data is skewed slightly right but mostly conforms to the shape of a normal curve.

Missing values:

Using `print(df.isnull().sum())`, we found that the columns had the following null value counts:

```
Unnamed: 0      0
Music           3
Techno          7
Movies          6
History         2
Mathematics     3
Pets            4
Spiders         5
Loneliness      1
Parents' advice 2
Internet usage  0
Finances        3
Age             7
Siblings        6
Gender          6
Village - town  4
```

Part 2: Exploratory Data Analysis (EDA)

2.1) UNIVARIATE ANALYSIS

Key Numerical Variables

Our dataset included the following numerical variables: `Movies`, `History`, `Mathematics`, `Pets`, `Spiders`, `Finances`, and `Age`.

The variables `Music`, `Parents' Advice`, `Techno`, and `Siblings` also classified as numerical variables, but we decided after some exploration and consideration that they were not key variables and decided to drop them (as is further detailed in section 3.1.1 of this document).

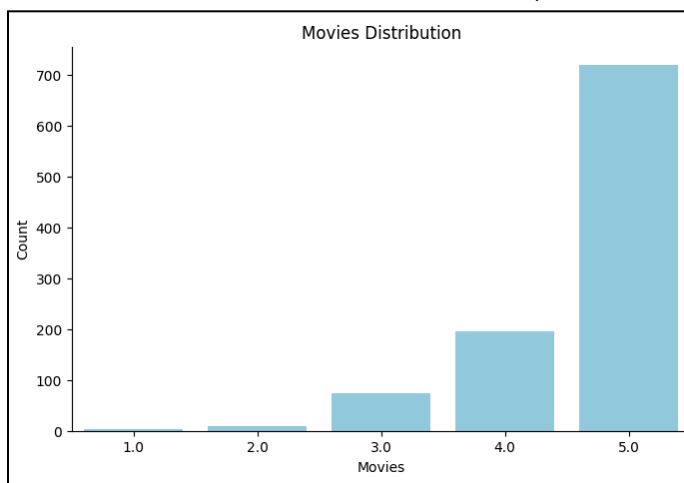


Figure 3: Movies Distribution Histogram - this variable has a heavy left skew, indicating that most participants liked movies.

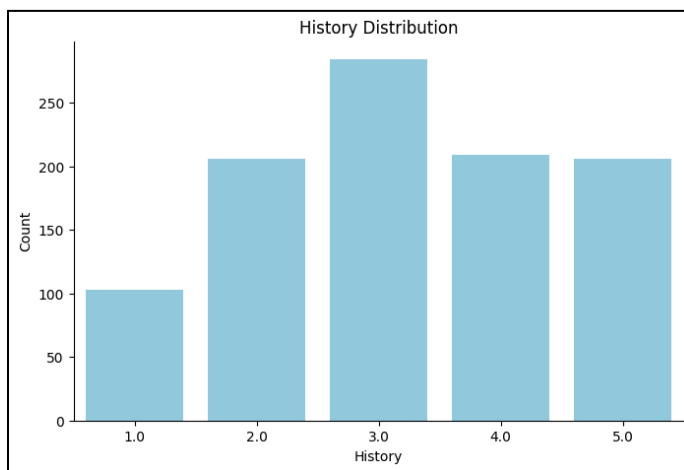


Figure 4: History Distribution Histogram - the distribution of liking for History appears to be approximately normal, with some slight right skew.

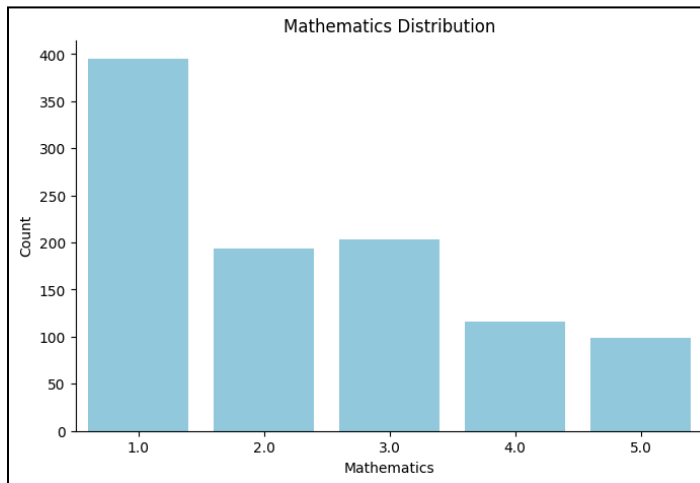


Figure 5: Mathematics Distribution Histogram - the liking for math, based on this distribution, is a lot more right-skewed and shows a trending dislike for math.

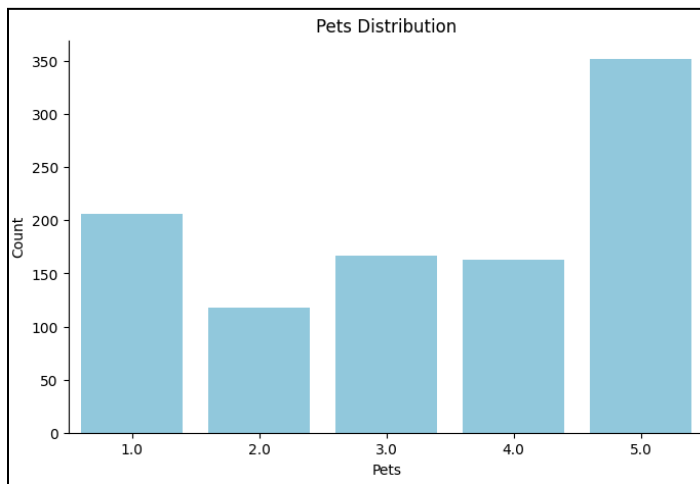


Figure 6: Pets Distribution Histogram - a high liking for pets creates a mode in this variable's distribution, while disliking pets creates a second, smaller mode. The bimodality of this variable may require special handling.

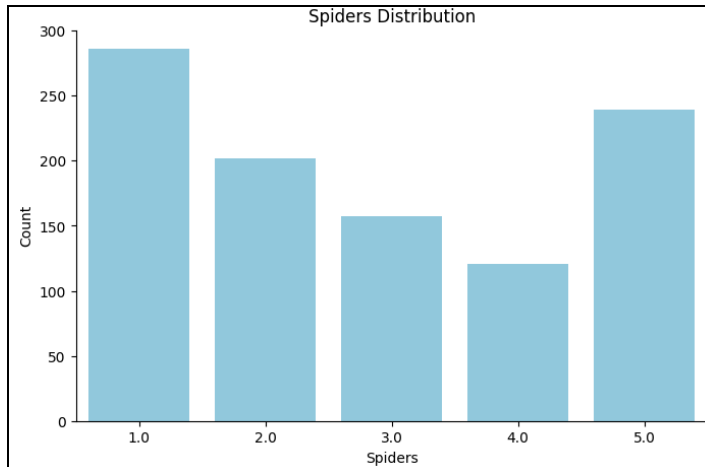


Figure 7: Spiders Histogram - similar to pets, this distribution is bimodal; but the larger mode clusters around disliking spiders rather than liking them. This variable may likewise require some specialized handling.

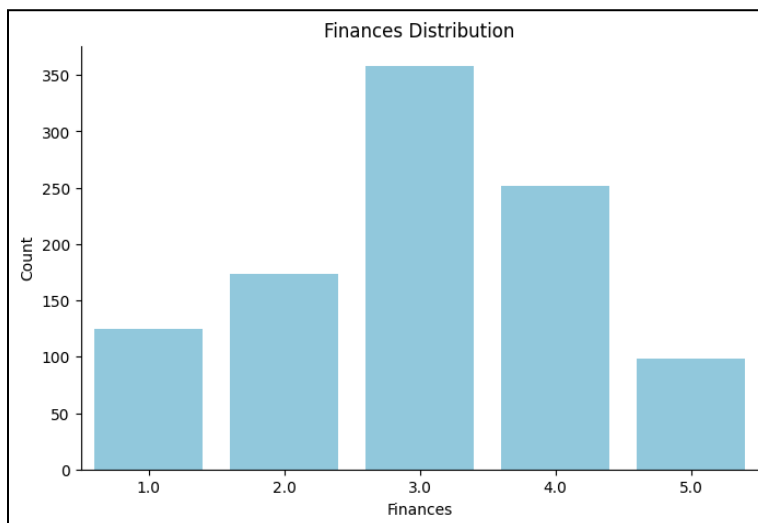


Figure 8: Finances Distribution Histogram - liking finances appears to be normally distributed, with most participants expressing neutral levels of like or dislike towards finances and the data skewing slightly left.

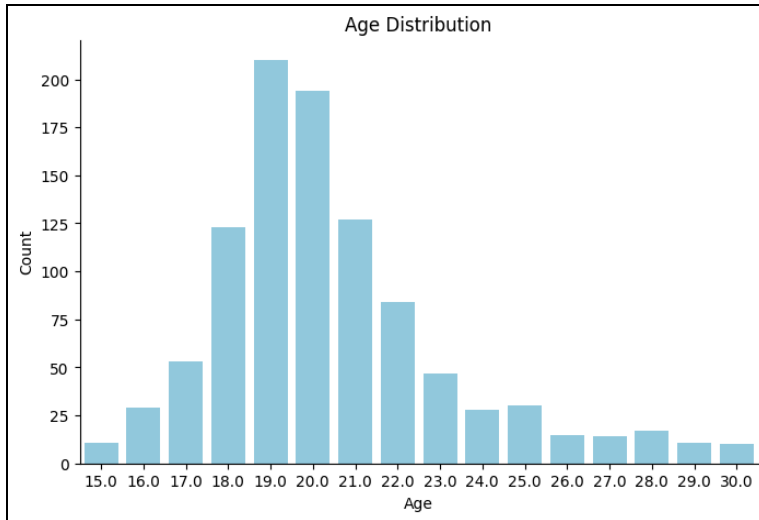


Figure 9: Age Distribution Histogram - age of participants appeared to be normally distributed, with the most popular age range landing around 19-20 and with a slight right skew.

Key Categorical Variables

The following figures show the frequency distributions for the categorical variables that we chose as prospective key variables (**Internet Usage**, **Gender**, and **Village - town**). There were no other categorical variables to consider in this dataset.

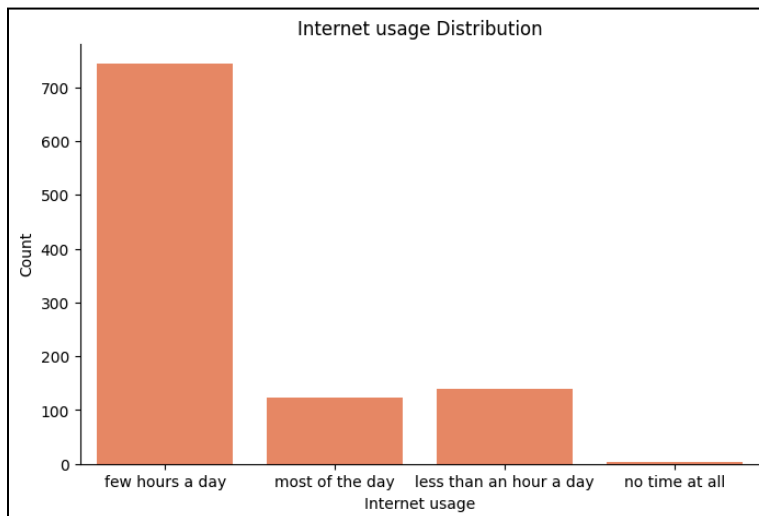


Figure 10: Internet Usage Distribution Bar Chart - we can see a strong imbalance of data in this variable trending towards most values being in the “few hours a day” category. A more specific breakdown of that range in the original data collection process may have yielded more precision in inferring any potentially dependent variables from this data.

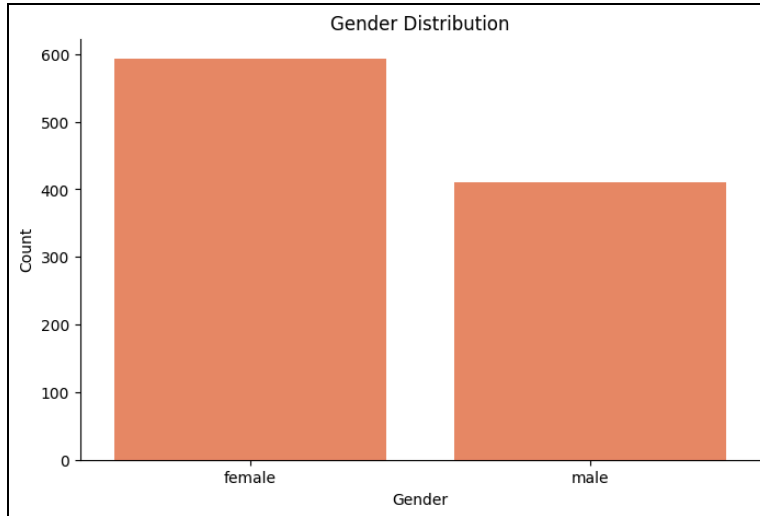


Figure 11: Gender Distribution Bar Chart - female respondents were around 33% more common than male in this dataset.

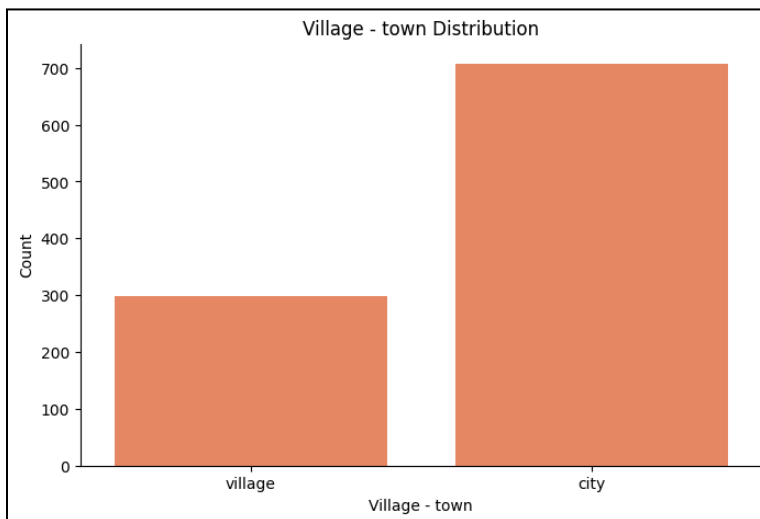


Figure 12: Village - Town Distribution Bar Chart - most participants (over 60%) indicated their residence in a city ("town") rather than a village.

2.2) RELATIONSHIP TO TARGET VARIABLE / MULTIVARIABLE ANALYSIS

The following graphs examine the relationships between the target variable (**Loneliness**) and the other variables of the dataset.

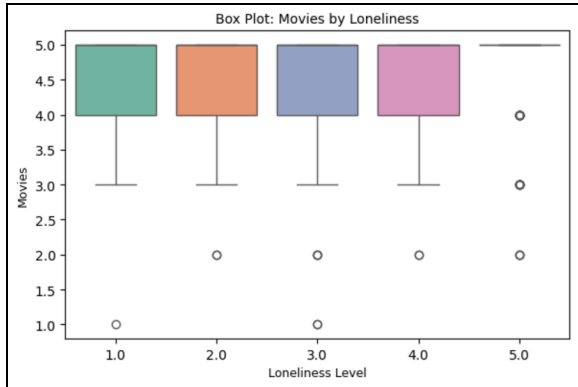


Figure 13: Box Plot of Movies by Loneliness
Based on this plot, there may be a slight positive correlation between strongly liking movies and experiencing loneliness.

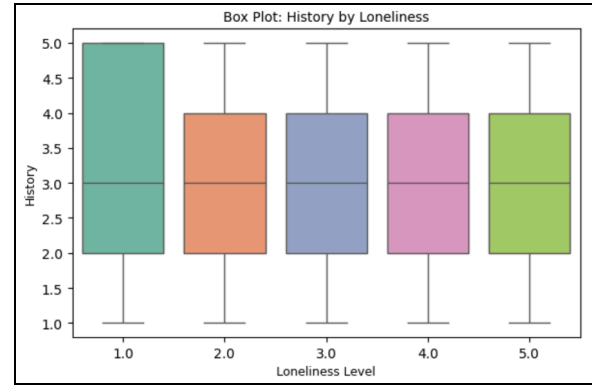


Figure 14: Box Plot of History by Loneliness
A possible slight negative correlation could be inferred from this plot between liking history and feeling lonely.

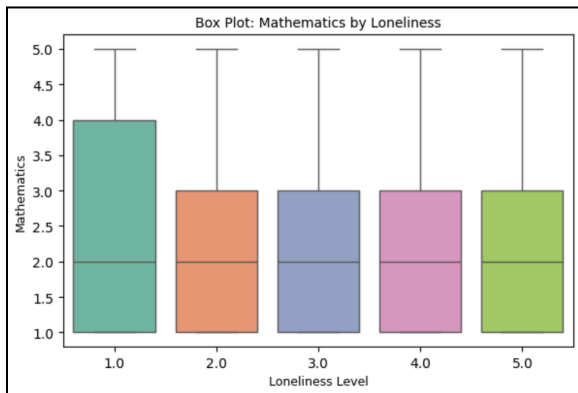


Figure 15: Box Plot of Mathematics by Loneliness - it appears that a weak, negative correlation (if any) is the only correlation between liking math and feeling loneliness.

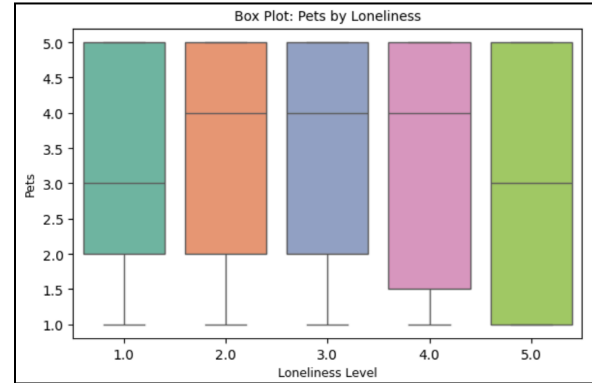


Figure 16: Box Plot of Pets by Loneliness - Correlations between loneliness and liking pets from the data in this plot are unclear or non-existent, due to irregular variance in the data.

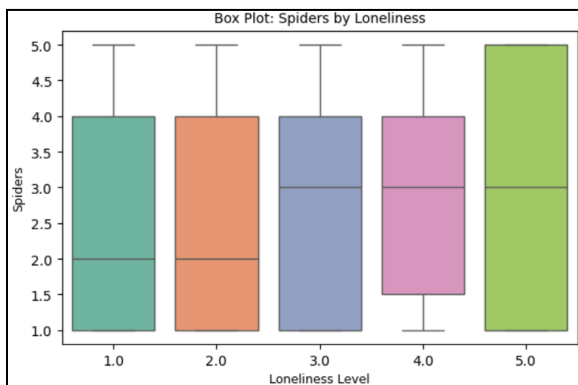


Figure 17: Box Plot of Spiders by Loneliness
A slight positive correlation could be inferred to exist between loneliness and liking spiders, based on this plot.

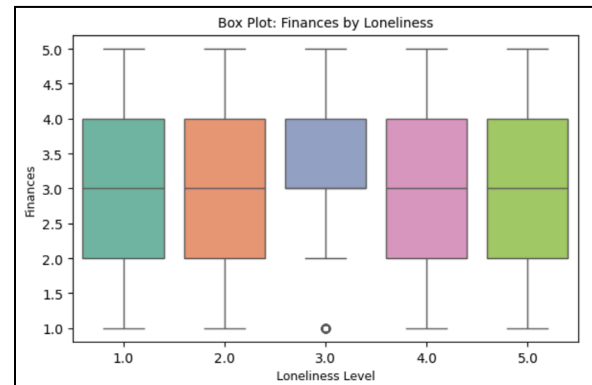


Figure 18: Box Plot of Finances by Loneliness - this plot does not offer any clear correlation data between liking finance and experiencing loneliness.

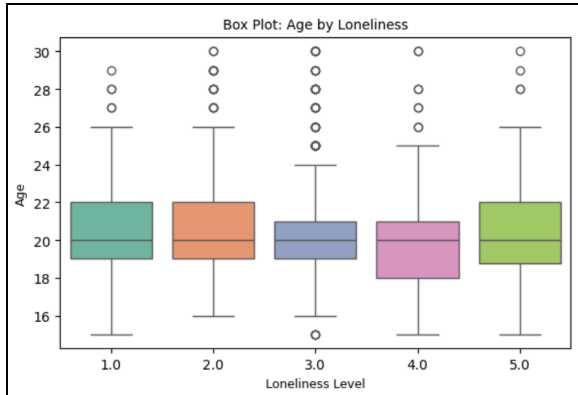


Figure 19: A box plot of Age by Loneliness. There does not appear to be a meaningful relationship between those

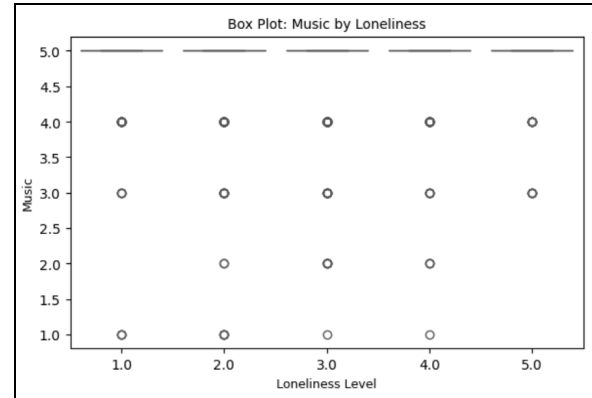


Figure 20: A box plot of music by loneliness. Nearly everyone likes music, so everything that isn't 5 is an outlier.

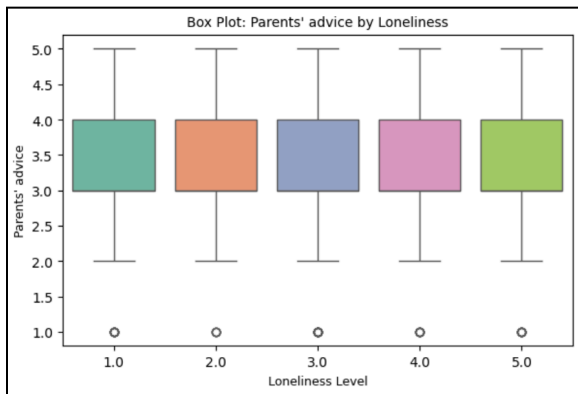


Figure 21: A box plot of Parents' advice by Loneliness. There does not appear to be a connection here between loneliness and techno as the plots are all the same

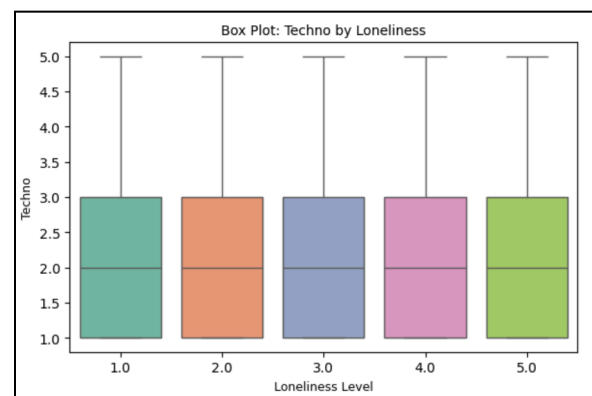


Figure 22: A box plot of Techno by loneliness. There does not appear to be a connection here between loneliness and techno as the plots are all the same

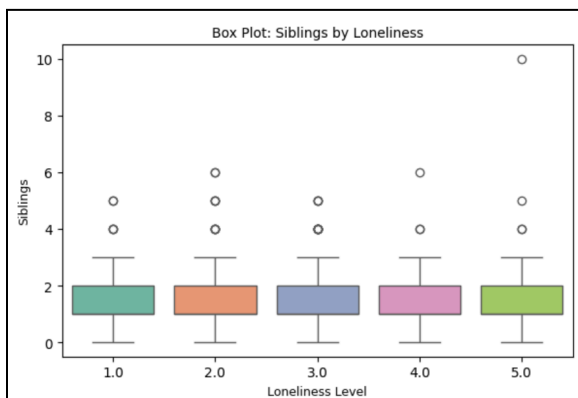


Figure 23: A box plot of siblings by loneliness. There does not appear to be a connection here between loneliness and techno as the plots are all the same

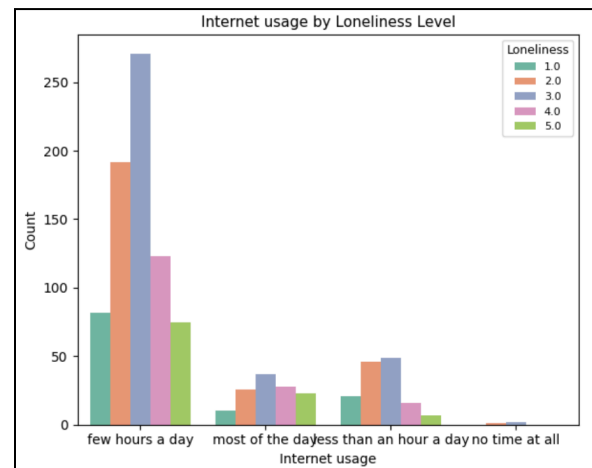


Figure 24: A count plot of internet usage by loneliness level.

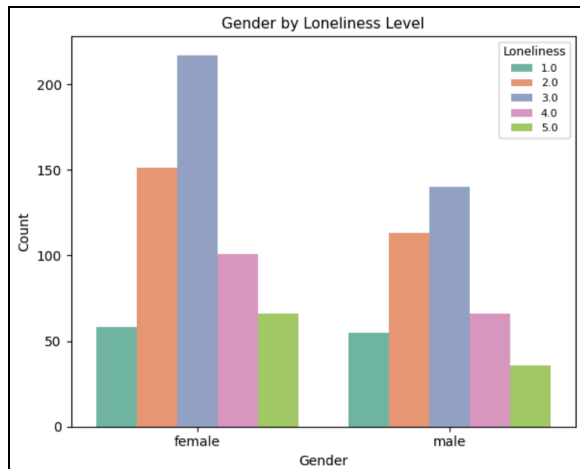


Figure 25: A count plot of gender by loneliness level.

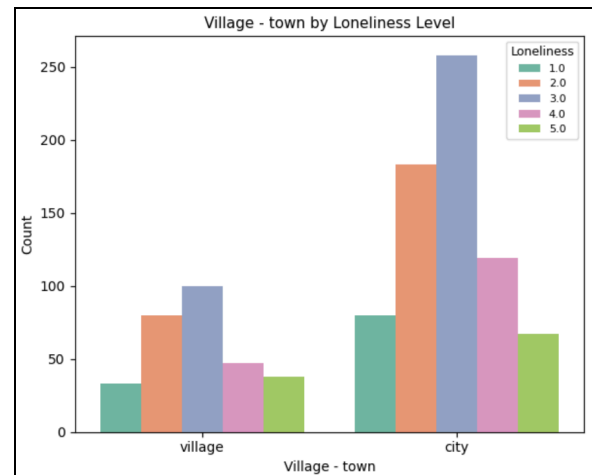


Figure 26: A count plot of village/city life by loneliness level.

2.3) CORRELATION ANALYSIS

There were no highly correlated variable pairs, and hence no multicollinearity concerns either. The key variables with the highest correlation to the target variable were Spiders and Age.

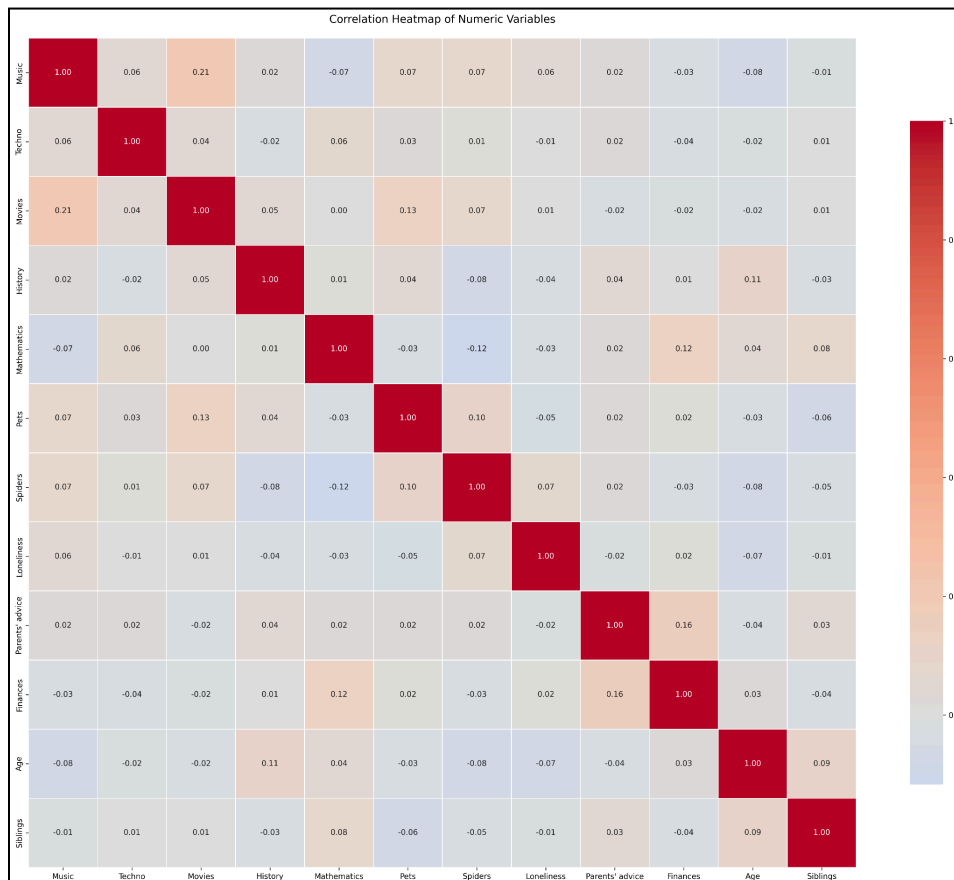


Figure 27: A heat map of the data correlations, demonstrating that they were very weak

2.4) KEY FINDINGS SUMMARY

There were no really hard-and-fast correlations in the dataset. What we were able to find were a few weak associations with loneliness. However, these only measured up to a 7% correlation at most. This was the correlation with spiders and loneliness. The rest didn't seem to have much impact on the prediction other than music, at 6%, and age, at -7%. (See figure 27)

We found it rather surprising and interesting that there was a correlation between Spiders and Loneliness. There was also a weaker correlation between pets and loneliness than we expected. We figured that there would be some, but it's very weak. We figured that it would be stronger than it was.

The only outlier that we really noticed was that there was one person with a sibling count of 10. One concern we had about the dataset was that there was very little correlation between any two variables. The heatmap that we did for correlation showed at the highest between any variables (not including our target) only a mere 12% correlation, which isn't very significant at all. This holds fairly consistent with common trends in psychological studies, especially when self-reported data is used.

Part 3: Data Preprocessing

3.1) FEATURE SELECTION

3.1.1) VARIABLE REMOVAL DOCUMENTATION

Exclude: Music, Parents' Advice, Techno, Siblings

Variable Name	Reason for Removal
Music	No relationship with the target variable, as per the EDA boxplot. (See figure 20)
Parents' Advice	No relationship between the two. Data stayed consistent between each categorization of loneliness.
Techno	This was not a well-understood variable as "Techno" could mean a few different things and was not clearly documented in the dataset. This variable also did not seem to have much correlation with the target variable.

Siblings	Siblings did not change for each of the categorizations of loneliness, thus having no significant correlation.
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3.1.2) FEATURE COUNT SUMMARY

There were a grand total of 1010 entries in this data set. Of those, we had 16 columns. This can be seen in figure 28.

RangeIndex: 1010 entries, 0 to 1009		
Data columns (total 16 columns):		
#	Column	Non-Null Count
0	Unnamed: 0	1010 non-null
1	Music	1007 non-null
2	Techno	1003 non-null
3	Movies	1004 non-null
4	History	1008 non-null
5	Mathematics	1007 non-null
6	Pets	1006 non-null
7	Spiders	1005 non-null
8	Loneliness	1009 non-null
9	Parents' advice	1008 non-null
10	Internet usage	1010 non-null
11	Finances	1007 non-null
12	Age	1003 non-null
13	Siblings	1004 non-null
14	Gender	1004 non-null
15	Village - town	1006 non-null

Figure 28: Feature count summary

3.2) HANDLING MISSING VALUES

3.2.1) MISSING DATA ANALYSIS

When it comes to how we are handling the missing values, we first need to see how much of the data is missing in each column. If too much is missing, we may need to take out the column. If there's enough data, we may be able to get away with just deleting the rows. If there's not enough to do that, we need to impute it. Looking at figure 29 we can see that there's at most ~0.7% missing data in any given column. Realistically, we could drop these rows and not lose any significant data, but we chose to impute them, taking the average, rounding to the nearest whole, and using that to impute the missing data.

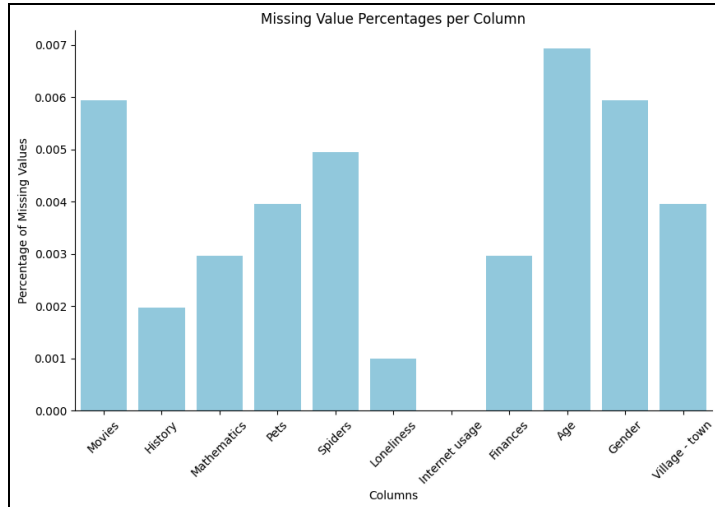


Figure 29: A bar chart showing where the data is missing from by degrees of percentages

3.2.2) REMOVAL DOCUMENTATION

The removed variables were those that when included in a boxplot, saw no real significance to the target variable. (See figure 30)

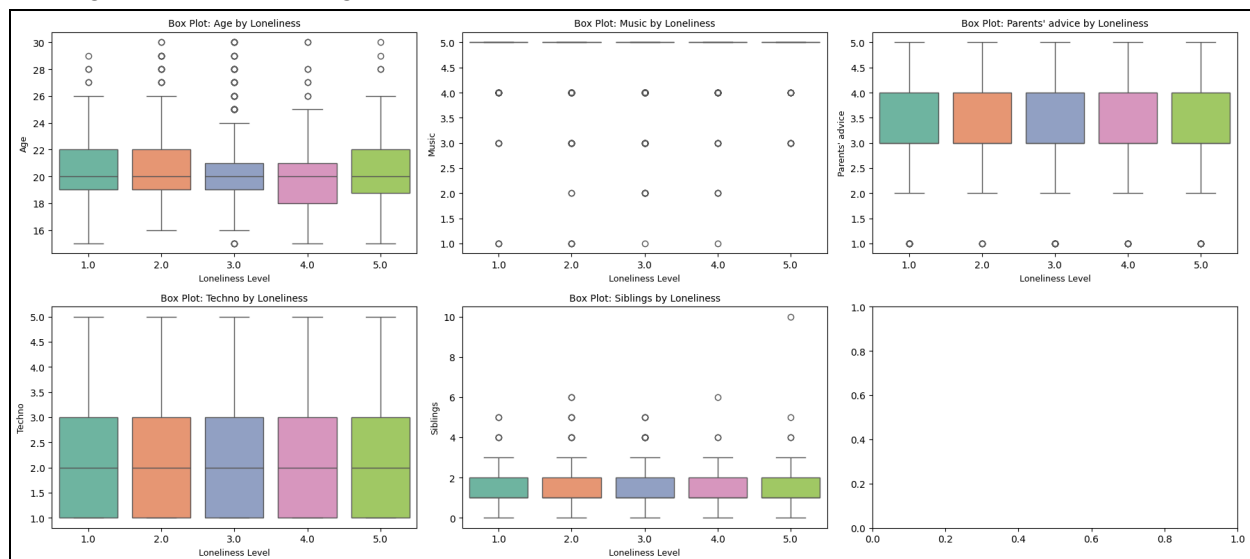


Figure 30: A box plot of the dropped data columns, with the exception of Age.

3.2.3) IMPUTATION STRATEGY TABLE

To impute this data, we first took the average of the column and rounded it to the nearest whole number, then we put that in for the missing data.

3.2.4) IMPUTATION RESULTS

The results were no more missing values and very little skewing of the data due to the averaging and rounding.

```

Before imputation:
Movies          5
History         1
Mathematics     2
Pets            3
Spiders         4
Loneliness      1
Internet usage  0
Finances        2
Age            6
Gender          5
Village - town  4
dtype: int64

After numerical imputation:
Movies          0
History         0
Mathematics     0
Pets            0
Spiders         0
Loneliness      0
Internet usage  0
Finances        0
Age            0
Gender          0
Village - town  0
dtype: int64

```

Figure 31: Imputation results

3.3) ENCODING CATEGORICAL VARIABLES

For the categorical variable of internet mapping, did an Ordinal Encoding. For Gender and Village - town we did One Hot Encoding.

3.3.1) CATEGORICAL VARIABLE IDENTIFICATION

The categorical variables are: ['Internet usage', 'Gender', 'Village - town']

The unique values in Internet usage are: ['few hours a day' 'most of the day' 'less than an hour a day'

'no time at all']

The unique values in Gender are: ['female' 'male']

The unique values in Village - town are: ['village' 'city']

3.3.2) ENCODING STRATEGY TABLE

The encoding strategy table is in figure 32. It goes over the strategy used and the justification. For Internet usage, it was an ordinal encoding with the justification of preserving order, or time spent. Gender was One-Hot Encoded with a binary nominal. Same for the Village - town column.

3.3.3) ENCODING RESULTS

After encoding the features into binaries and counted variables, we are left with a very clean dataset that is ready for training. See figure 32.

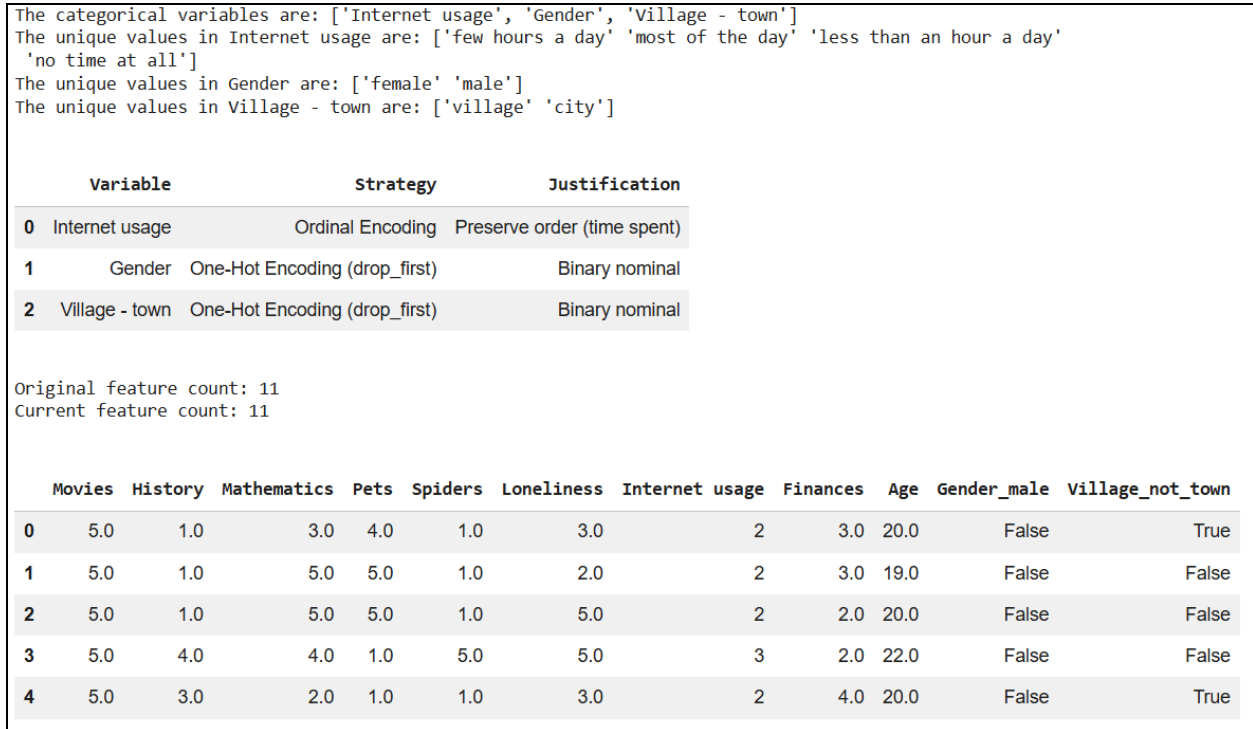


Figure 32: Encoding results

Part 4: Cross Validation (Train-Test Split)

4.1) SPLIT STRATEGY

We split the data with an 80:20 spread. This allowed us ample training and testing data to run with.

4.2) DATASET DIMENSIONS TABLE

The data set dimensions table can be found in figure 33. It shows that we have 804 samples in training and 202 in testing. There are 11 features that are a part of the training set providing ample training for the model.

4.3) TARGET DISTRIBUTION ANALYSIS

Our target distribution analysis bar chart can be found in figure 33 showing a very similar distribution of loneliness in both the training and testing data sets. This ensures that our model won't be confused or have skewed results due to a skew in the data being trained and tested on.

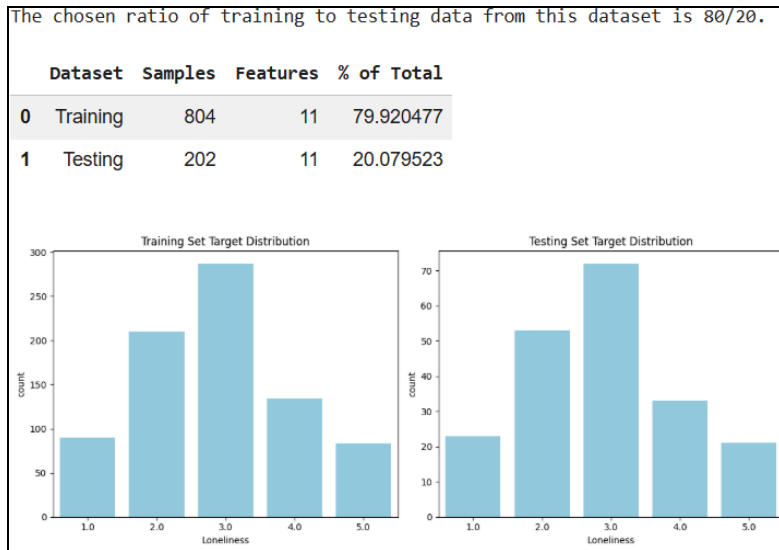


Figure 33: A distribution of the testing and training data sets

Part 5: Feature Scaling

5.1) SCALING DECISION TABLE

Scaling Strategy:				
	Variable Group	Original Range	Strategy	Justification
0	Age	15-30	StandardScaler	Standardize to Mean=0, Std=1
1	Likert Features (1-5)	1-5	StandardScaler	Standardize to Mean=0, Std=1
2	Internet Usage (Ordinal)	0-3	StandardScaler	Standardize to Mean=0, Std=1
3	Binary Variables	0-1	StandardScaler	Standardize to Mean=0, Std=1

Figure 34: Scaling decision table

5.2) SCALING METHODOLOGY

```
features = [col for col in train_df.columns if col != 'Loneliness']
scaler = StandardScaler()
train_df[features] = scaler.fit_transform(train_df[features])
test_df[features] = scaler.transform(test_df[features])
```

Figure 35: Code for the scaling methodology

- I ran `fit_transform` on `train_df`, and `transform` on `test_df`, which confirms that I did the fitting on the training data only.
- When fitting using `StandardScaler`, it learns the means and standard deviations of all of the features to standardize them. Testing data is meant to represent completely unseen

data that can be used to evaluate the accuracy of a model. Fitting on testing data completely defeats the purpose and makes such an evaluation unreliable.

5.3) SCALING VERIFICATION

Training data after scaling:											
	Movies	History	Mathematics	Pets	Spiders	Loneliness	Internet usage	Finances	Age	Gender_male	Village_not_town
count	804.00	804.00	804.00	804.00	804.00	804.00	804.00	804.00	804.00	804.00	804.00
mean	-0.00	-0.00	0.00	-0.00	-0.00	2.89	-0.00	0.00	0.00	-0.00	0.00
std	1.00	1.00	1.00	1.00	1.00	1.13	1.00	1.00	1.00	1.00	1.00
min	-5.18	-1.74	-0.97	-1.54	-1.21	1.00	-3.83	-1.77	-1.93	-0.83	-0.65
25%	-0.89	-0.94	-0.97	-0.89	-1.21	2.00	0.04	-0.89	-0.49	-0.83	-0.65
50%	0.55	-0.14	-0.22	0.42	0.09	3.00	0.04	-0.02	-0.13	-0.83	-0.65
75%	0.55	0.66	0.54	1.07	0.74	4.00	0.04	0.86	0.24	1.21	1.54
max	0.55	1.46	2.05	1.07	1.39	5.00	1.98	1.73	3.49	1.21	1.54

Testing data after scaling:											
	Movies	History	Mathematics	Pets	Spiders	Loneliness	Internet usage	Finances	Age	Gender_male	Village_not_town
count	202.00	202.00	202.00	202.00	202.00	202.00	202.00	202.00	202.00	202.00	202.00
mean	-0.01	0.10	0.18	-0.08	-0.11	2.88	0.00	0.02	0.14	0.01	-0.01
std	0.94	1.05	1.09	1.02	0.99	1.14	1.04	1.00	1.08	1.00	1.00
min	-3.75	-1.74	-0.97	-1.54	-1.21	1.00	-3.83	-1.77	-1.93	-0.83	-0.65
25%	-0.89	-0.94	-0.97	-0.89	-1.21	2.00	0.04	-0.89	-0.49	-0.83	-0.65
50%	0.55	-0.14	-0.22	-0.23	-0.56	3.00	0.04	-0.02	-0.13	-0.83	-0.65
75%	0.55	0.66	1.29	1.07	0.74	4.00	0.04	0.86	0.60	1.21	1.54
max	0.55	1.46	2.05	1.07	1.39	5.00	1.98	1.73	3.49	1.21	1.54

Figure 36: Verification table for the scaling

5.4) VISUAL COMPARISON

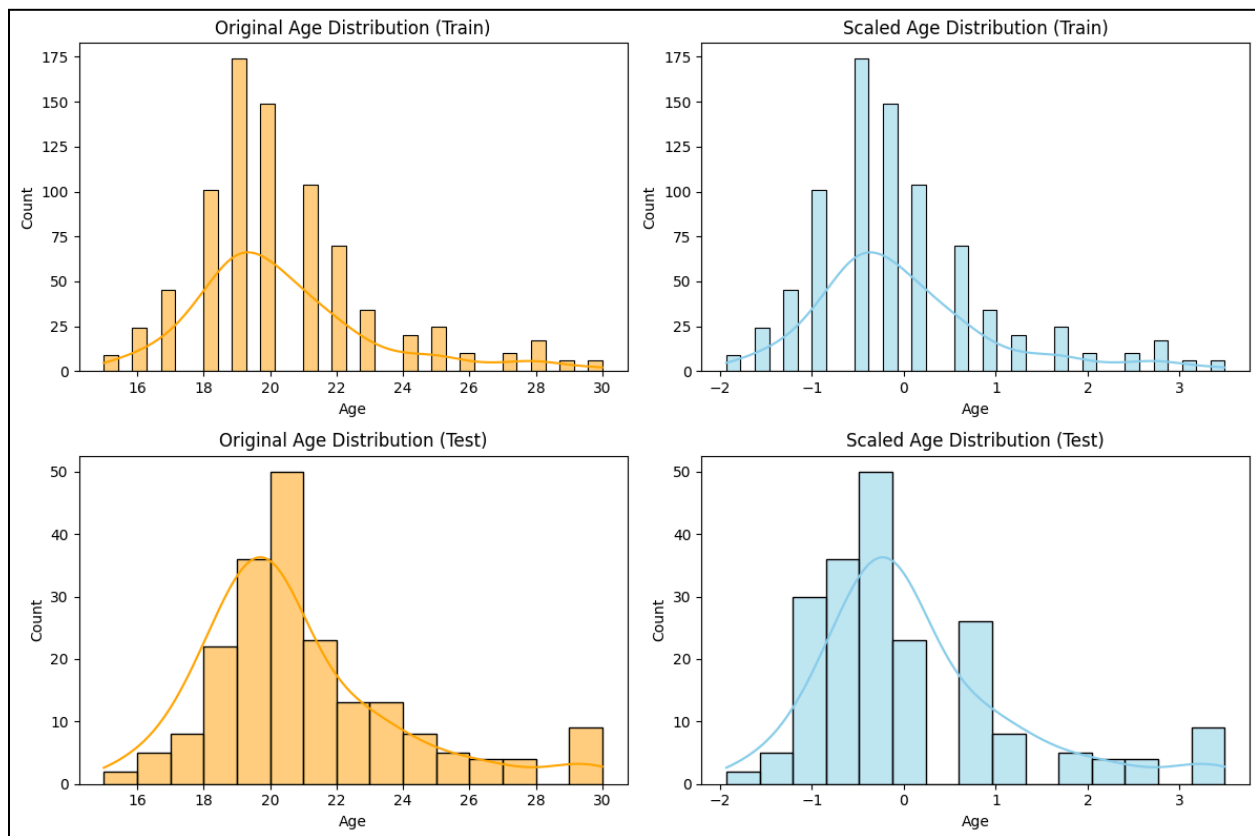


Figure 37: A visual comparison of the distribution of the training and test datasets before and after scaling, showing that they still have the same shape.

Part 6: Validation (didn't do)

Part 7: Summary and Reflection

7.1) PROCESSING SUMMARY TABLE

Preprocessing Step	Initial	Final	Change
Samples	1010	1006	-4
Features	16	11	-5
Missing Values	59	0	All Handled

7.2) KEY INSIGHTS

- Most important findings from EDA
 - None of the variables are highly correlated with each other.
 - We made assumptions about what the relationships between the variables would be, but none of them were correct.
- Most challenging aspect of preprocessing
 - For us, the most challenging aspect of preprocessing was determining how we should impute the missing values in the rows and columns of this dataset. We had to study the data and gain more contextual knowledge to do this.
- Variables with most missing data
 - The Age, Gender, Movies, and Spiders columns had the most missing data.
 - Out of over 1000 observations, no column was missing more than 7 values.
- New features from one-hot encoding
 - We did not add any new features through one-hot encoding. Out of the three categorical variables that we identified, one of them was converted to a numeric variable, and the other two were converted to binary variables.
- Any concerns about final dataset
 - It will likely be very hard to find a meaningful relationship between any of these variables. We did not notice any strong patterns.
- How EDA informed your preprocessing decisions
 - When we were following the EDA process, we found that most of the columns were discrete numeric data types. We also found multiple columns that represented very similar things or provided meaningless data. We were able to remove these columns early on.
 - Our figures and graphs gave us insight into how we should impute and scale the data. We picked standard scaling because most columns followed a normal distribution.